**1) a)**
Rotation, anticlockwise about the origin, by $90°$
**Alternative Method:**
Rotation, clockwise about the origin, by $270°$

**1) b)**
Rotation, about the origin, by $180°$

**1) c)**
Rotation, anticlockwise about the origin, by $270°$
**Alternative Method:**
Rotation, clockwise about the origin, by $90°$

**2) a)**
$\mathbf{A} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

**2) b)**
$\mathbf{A}^2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$

**2) c)**
$\mathbf{A}^3 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}^3 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$

**2) d)**
$\mathbf{A}^4 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}^4 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

**2) e)**
$\mathbf{A}^5 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}^5 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

**2) f)**
$\mathbf{A}^{2025} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}^{2025} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

**3)**
This is a Rotation, anticlockwise about the origin, by $90°$, so area is preserved.
Hence the area of the shape's image is $4$ units$^2$

**4)**

$\begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix}\begin{pmatrix}6\\ -2\end{pmatrix}=\begin{pmatrix}2\\ 6\end{pmatrix}$

So the coordinates of $B$ are $(2,6)$

**5)**

$\begin{pmatrix}-1 & 0\\ 0 & -1\end{pmatrix}\begin{pmatrix}\sqrt{2}\\ -2\sqrt{2}\end{pmatrix}=\begin{pmatrix}-\sqrt{2}\\ 2\sqrt{2}\end{pmatrix}$

So the coordinates of $B$ are $\left(-\sqrt{2},2\sqrt{2}\right)$

**6)**

$\frac{990°}{90°}=11$

$\frac{11}{4}=2\text{ r }3$

A rotation, anticlockwise about the origin, by $90°$ can be represented by the matrix:

$\begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix}$

$\begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix}^2=\begin{pmatrix}-1 & 0\\ 0 & -1\end{pmatrix}$

$\begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix}^3=\begin{pmatrix}0 & 1\\ -1 & 0\end{pmatrix}$

$\begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix}^4=\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}$

$\begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix}^5=\begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix}$

…

So $\begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix}^{11}=\begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix}^3=\begin{pmatrix}0 & 1\\ -1 & 0\end{pmatrix}$

$\begin{pmatrix}0 & 1\\ -1 & 0\end{pmatrix}\begin{pmatrix}5\\ -17\end{pmatrix}=\begin{pmatrix}-17\\ -5\end{pmatrix}$

So the coordinates of $B$ are $(-17,-5)$

**7)**

$\begin{pmatrix}0 & 1\\ -1 & 0\end{pmatrix}\begin{pmatrix}1 & 2 & 6\\ 3 & 8 & -1\end{pmatrix}=\begin{pmatrix}3 & 8 & -1\\ -1 & -2 & -6\end{pmatrix}$

So the coordinates of the vertices of $T'$ are $(3,-1)$, $(8,-2)$ and $(-1,-6)$

**8)**

$\begin{pmatrix}-1 & 0\\ 0 & -1\end{pmatrix}\begin{pmatrix}0 & 0 & 10 & 10\\ 2 & 6 & 6 & 2\end{pmatrix}=\begin{pmatrix}0 & 0 & -10 & -10\\ -2 & -6 & -6 & -2\end{pmatrix}$

So the coordinates of the vertices of $R'$ are $(0,-2)$, $(0,-6)$, $(-10,-6)$ and $(-10,-2)$

**9) a)**

$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 2 \\ -5 \end{pmatrix} = \begin{pmatrix} 5 \\ 2 \end{pmatrix}$

A rotation, anticlockwise about the origin, by $90°$

**9) b)**

$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

**9) c)**

Distance from $(2, -5)$ to the origin is $\sqrt{2^2 + (-5)^2} = \sqrt{29}$

Length of arc $= \frac{90}{360} \times 2\pi \times \sqrt{29} = \frac{\pi\sqrt{29}}{2}$ units

**10)**

The transformation appears to be a rotation, clockwise about the origin, by $90°$ (or a rotation, anticlockwise about the origin, by $270°$):

$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} -2 & -5 & -5 & -2 \\ 2 & 4 & 6 & 4 \end{pmatrix} = \begin{pmatrix} a & b & c & 4 \\ d & e & f & 2 \end{pmatrix}$

$\Rightarrow \begin{pmatrix} 2 & 4 & 6 & 4 \\ 2 & 5 & 5 & 2 \end{pmatrix} = \begin{pmatrix} a & b & c & 4 \\ d & e & f & 2 \end{pmatrix}$

So the coordinates of the vertices of the image are $(2,2)$, $(4,5)$, $(6,5)$ and $(4,2)$

$\therefore \mathbf{M} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ and $A'(2,2)$, $B'(4,5)$, $C'(6,5)$

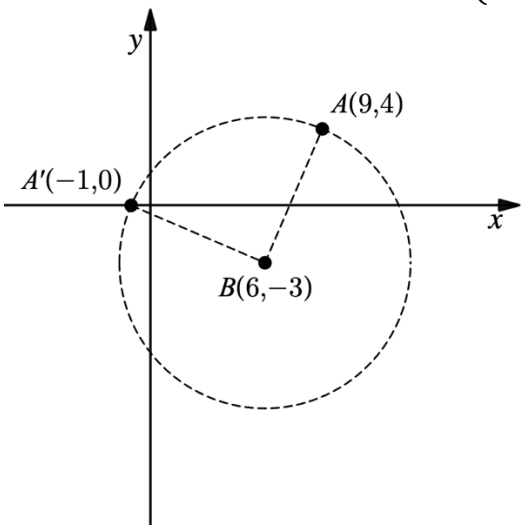
**11)**

$\overrightarrow{BA} = \overrightarrow{OA} - \overrightarrow{OB} = \begin{pmatrix} 9 \\ 4 \end{pmatrix} - \begin{pmatrix} 6 \\ -3 \end{pmatrix} = \begin{pmatrix} 3 \\ 7 \end{pmatrix}$

$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 3 \\ 7 \end{pmatrix} = \begin{pmatrix} -7 \\ 3 \end{pmatrix}$

$\overrightarrow{OA'} = \overrightarrow{OB} + \begin{pmatrix} -7 \\ 3 \end{pmatrix} = \begin{pmatrix} 6 \\ -3 \end{pmatrix} + \begin{pmatrix} -7 \\ 3 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$

So the coordinates of $A'$ are $(-1, 0)$




**12) a)**

$$\begin{pmatrix} \cos 45° & -\sin 45° \\ \sin 45° & \cos 45° \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$$

**12) b)**

A rotation by $45°$, clockwise about the origin is equivalent to a rotation by $315°$, anticlockwise about the origin.

$$\begin{pmatrix} \cos 315° & -\sin 315° \\ \sin 315° & \cos 315° \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$$

**12) c)**

$$\begin{pmatrix} \cos 135° & -\sin 135° \\ \sin 135° & \cos 135° \end{pmatrix} = \begin{pmatrix} -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix}$$

**12) d)**

A rotation by 135°, clockwise about the origin is equivalent to a rotation by 225°, anticlockwise about the origin.

$$\begin{pmatrix} \cos 225° & -\sin 225° \\ \sin 225° & \cos 225° \end{pmatrix} = \begin{pmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix}$$

**13)**

$$\begin{pmatrix} \cos 45° & -\sin 45° \\ \sin 45° & \cos 45° \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$$

$$\begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} \begin{pmatrix} -2 & 8 & 4 \\ 5 & 3 & -2 \end{pmatrix} = \begin{pmatrix} -\frac{7\sqrt{2}}{2} & \frac{5\sqrt{2}}{2} & 3\sqrt{2} \\ \frac{3\sqrt{2}}{2} & \frac{11\sqrt{2}}{2} & \sqrt{2} \end{pmatrix}$$

So the coordinates of the vertices of $T'$ are $\left(-\frac{7\sqrt{2}}{2}, \frac{3\sqrt{2}}{2}\right)$, $\left(\frac{5\sqrt{2}}{2}, \frac{11\sqrt{2}}{2}\right)$ and $(3\sqrt{2}, \sqrt{2})$

**14)**

$360° - 225° = 135°$

$$\begin{pmatrix} \cos 135° & -\sin 135° \\ \sin 135° & \cos 135° \end{pmatrix} = \begin{pmatrix} -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix}$$

$$\begin{pmatrix} -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix} \begin{pmatrix} -3 & -3 & 6 & 6 \\ 2 & 4 & 4 & 2 \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & -5\sqrt{2} & -4\sqrt{2} \\ -\frac{5\sqrt{2}}{2} & -\frac{7\sqrt{2}}{2} & \sqrt{2} & 2\sqrt{2} \end{pmatrix}$$

So the coordinates of the vertices of $R'$ are $\left(\frac{\sqrt{2}}{2}, -\frac{5\sqrt{2}}{2}\right)$, $\left(-\frac{\sqrt{2}}{2}, -\frac{7\sqrt{2}}{2}\right)$, $(-5\sqrt{2}, \sqrt{2})$ and $(-4\sqrt{2}, 2\sqrt{2})$

**15)**

$$\begin{pmatrix} \cos 135° & -\sin 135° \\ \sin 135° & \cos 135° \end{pmatrix} = \begin{pmatrix} -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix}$$

$$\begin{pmatrix} -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix} \begin{pmatrix} -\sqrt{2} & k\sqrt{2} & 1 \\ 3\sqrt{2} & 5\sqrt{2} & -1 \end{pmatrix} = \begin{pmatrix} -2 & -13 & 0 \\ -4 & 3 & \sqrt{2} \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} -2 & -k-5 & 0 \\ -4 & k-5 & \sqrt{2} \end{pmatrix} = \begin{pmatrix} -2 & -13 & 0 \\ -4 & 3 & \sqrt{2} \end{pmatrix}$$

$\therefore k - 5 = 3 \Rightarrow k = 8$

**16) a)**

$$\begin{pmatrix} \cos 30° & -\sin 30° \\ \sin 30° & \cos 30° \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$$

**16) b)**

$$\begin{pmatrix} \cos 60° & -\sin 60° \\ \sin 60° & \cos 60° \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$$

**16) c)**

$$\begin{pmatrix} \cos 120° & -\sin 120° \\ \sin 120° & \cos 120° \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$$

**16) d)**

$$\begin{pmatrix} \cos 150° & -\sin 150° \\ \sin 150° & \cos 150° \end{pmatrix} = \begin{pmatrix} -\frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{\sqrt{3}}{2} \end{pmatrix}$$

**16) e)**

A rotation by $210°$, clockwise about the origin, is the same as
a rotation by $360° - 210° = 150°$, anti-clockwise about the origin:

$$\begin{pmatrix} \cos 150° & -\sin 150° \\ \sin 150° & \cos 150° \end{pmatrix} = \begin{pmatrix} -\frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{\sqrt{3}}{2} \end{pmatrix}$$

**17)**

$$\begin{pmatrix} \cos 18° & -\sin 18° \\ \sin 18° & \cos 18° \end{pmatrix} = \begin{pmatrix} 0.951 & -0.309 \\ 0.309 & 0.951 \end{pmatrix}$$ (3 SF)

**18)**

$360° - 25° = 335°$

$$\begin{pmatrix} \cos 335° & -\sin 335° \\ \sin 335° & \cos 335° \end{pmatrix} = \begin{pmatrix} 0.906 & 0.423 \\ -0.423 & 0.906 \end{pmatrix}$$ (3 SF)

**19) a)**

$\cos\theta = -\frac{1}{2} \Rightarrow \theta = \{120°, 240°\}$

$\sin\theta = \frac{\sqrt{3}}{2} \Rightarrow \theta = \{60°, 120°\}$

Rotation, anticlockwise about the origin, by $120°$

**19) b)**

$\cos\theta = -\frac{\sqrt{3}}{2} \Rightarrow \theta = \{150°, 210°\}$

$\sin\theta = -\frac{1}{2} \Rightarrow \theta = \{210°, 330°\}$

Rotation, anticlockwise about the origin, by $210°$

**19) c)**

$\cos\theta = -\frac{\sqrt{2}}{2} \Rightarrow \theta = \{135°, 225°\}$

$\sin\theta = \frac{\sqrt{2}}{2} \Rightarrow \theta = \{45°, 135°\}$

Rotation, anticlockwise about the origin, by $135°$

**19) d)**

$\cos\theta = \frac{\sqrt{10+2\sqrt{5}}}{4} \Rightarrow \theta = \{18°, 342°\}$

$\sin\theta = \frac{-1+\sqrt{5}}{4} \Rightarrow \theta = \{18°, 162°\}$

Rotation, anticlockwise about the origin, by $18°$

**20)**

$\cos\theta = \frac{5}{\sqrt{41}} \Rightarrow \theta = \{38.7°, 321°\}$ (3 SF)

$\sin\theta = \frac{4}{\sqrt{41}} \Rightarrow \theta = \{38.7°, 141°\}$ (3 SF)

So $\theta = 38.7°$ (3 SF)

**21)**

$\cos\theta = \frac{\sqrt{6}-\sqrt{2}}{4} \Rightarrow \theta = \{75°, 285°\}$

$\sin\theta = -\frac{\sqrt{6}+\sqrt{2}}{4} \Rightarrow \theta = \{255°, 285°\}$

So $\theta = 285°$

**22) a)**

$\cos\theta = \frac{13}{85} \Rightarrow \theta = \{81.2°, 279°\}$ (3 SF)

$\sin\theta = \frac{84}{85} \Rightarrow \theta = \{81.2°, 98.8°\}$ (3 SF)

So yes, this represents a rotation, anticlockwise about the origin, by $81.2°$ (3 SF)

**22) b)**

$\cos\theta = -\frac{13}{85} \Rightarrow \theta = \{98.8°, 261°\}$ (3 SF)

$\cos\theta = \frac{13}{85} \Rightarrow \theta = \{81.2°, 279°\}$ (3 SF)

$\sin\theta = \frac{84}{85} \Rightarrow \theta = \{81.2°, 98.8°\}$ (3 SF)

So no, this does not represent a rotation as there is no common angle in the solutions to the three equations.

**22) c)**

$\cos\theta = \frac{13}{85} \Rightarrow \theta = \{81.2°, 279°\}$ (3 SF)

$\sin\theta = -\frac{84}{85} \Rightarrow \theta = \{261°, 279°\}$ (3 SF)

So yes, this represents a rotation, anticlockwise about the origin, by $279°$ (3 SF)

**22) d)**

$\cos\theta = \frac{84}{85} \Rightarrow \theta = \{8.80°, 351°\}$ (3 SF)

$\sin\theta = \frac{13}{85} \Rightarrow \theta = \{8.80°, 171°\}$ (3 SF)

So yes, this represents a rotation, anticlockwise about the origin, by $8.80°$ (3 SF)

**23)**

$\begin{pmatrix} a & -b \\ b & a \end{pmatrix}\begin{pmatrix} 2 \\ 10 \end{pmatrix} = \begin{pmatrix} -10 \\ -2 \end{pmatrix} \Rightarrow \begin{pmatrix} 2a - 10b \\ 2b + 10a \end{pmatrix} = \begin{pmatrix} -10 \\ -2 \end{pmatrix}$

$2a - 10b = -10$ (1)

$10a + 2b = -2$ (2)

$\therefore a = -\frac{5}{13}$ and $b = \frac{12}{13}$

$\cos\theta = -\frac{5}{13} \Rightarrow \theta = \{113°, 247°\}$ (3 SF)

$\sin\theta = \frac{12}{13} \Rightarrow \theta = \{67.4°, 113°\}$ (3 SF)

So $\theta = 113°$ to 3sf

**24)**

$\begin{pmatrix} a & -b \\ b & a \end{pmatrix}\begin{pmatrix} 15 \\ -2 \end{pmatrix} = \begin{pmatrix} 2 \\ -15 \end{pmatrix} \Rightarrow \begin{pmatrix} 15a+2b \\ 15b-2a \end{pmatrix} = \begin{pmatrix} 2 \\ -15 \end{pmatrix}$

$15a + 2b = 2$ (1)

$-2a + 15b = -15$ (2)

$\therefore a = \frac{60}{229}$ and $b = -\frac{221}{229}$

$\cos\theta = \frac{60}{229} \Rightarrow \theta = \{74.8°, 285°\}$ (3 SF)

$\sin\theta = -\frac{221}{229} \Rightarrow \theta = \{255°, 285°\}$ (3 SF)

So $\theta = 285°$ to 3sf

**25)**

$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}\begin{pmatrix} 1 \\ 5 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ \frac{\sqrt{103}}{2} \end{pmatrix}$

$\cos\theta - 5\sin\theta = \frac{1}{2} \Rightarrow \cos\theta = \frac{1}{2} + 5\sin\theta$ (1)

$\sin\theta + 5\cos\theta = \frac{\sqrt{103}}{2}$ (2)

Substitute (1) into (2):

$\sin\theta + 5\left(\frac{1}{2} + 5\sin\theta\right) = \frac{\sqrt{103}}{2}$

$\Rightarrow \sin\theta + \frac{5}{2} + 25\sin\theta = \frac{\sqrt{103}}{2}$

$\Rightarrow 26\sin\theta = \frac{\sqrt{103}}{2} - \frac{5}{2}$

$\Rightarrow \sin\theta = \frac{\sqrt{103}}{52} - \frac{5}{52} \Rightarrow \theta = \{5.68°, 174°\}$ (3 SF)

$\cos\theta = \frac{1}{2} + 5\left(\frac{\sqrt{103}}{52} - \frac{5}{52}\right) = \frac{1}{52} + \frac{5\sqrt{103}}{52} \Rightarrow \theta = \{5.68°, 354°\}$ (3 SF)

$\mathbf{M} = \begin{pmatrix} \frac{1}{52} + \frac{5\sqrt{103}}{52} & \frac{5}{52} - \frac{\sqrt{103}}{52} \\ \frac{\sqrt{103}}{52} - \frac{5}{52} & \frac{1}{52} + \frac{5\sqrt{103}}{52} \end{pmatrix} = \begin{pmatrix} 0.995 & -0.099 \\ 0.099 & 0.995 \end{pmatrix}$ (3 DP)

$\begin{pmatrix} \frac{1}{52} + \frac{5\sqrt{103}}{52} & \frac{5}{52} - \frac{\sqrt{103}}{52} \\ \frac{\sqrt{103}}{52} - \frac{5}{52} & \frac{1}{52} + \frac{5\sqrt{103}}{52} \end{pmatrix}\begin{pmatrix} 3 & 4 & 1 \\ 1 & 2 & 5 \end{pmatrix} = \begin{pmatrix} \frac{4+7\sqrt{103}}{26} & \frac{7+9\sqrt{103}}{26} & \frac{1}{2} \\ \frac{-7+4\sqrt{103}}{26} & \frac{-9+7\sqrt{103}}{26} & \frac{\sqrt{103}}{2} \end{pmatrix}$

So $A'\left(\frac{4+7\sqrt{103}}{26}, \frac{-7+4\sqrt{103}}{26}\right) = A'(2.886, 1.292)$ (3 DP)

$B'\left(\frac{7+9\sqrt{103}}{26}, \frac{-9+7\sqrt{103}}{26}\right) = B'(3.782, 2.386)$ (3 DP)

With $\mathbf{M} = \begin{pmatrix} \frac{1}{52} + \frac{5\sqrt{103}}{52} & \frac{5}{52} - \frac{\sqrt{103}}{52} \\ \frac{\sqrt{103}}{52} - \frac{5}{52} & \frac{1}{52} + \frac{5\sqrt{103}}{52} \end{pmatrix} = \begin{pmatrix} 0.995 & -0.099 \\ 0.099 & 0.995 \end{pmatrix}$ (3 DP)

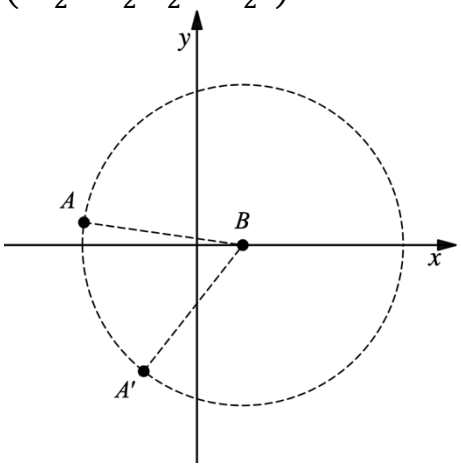
**26)**

$\overrightarrow{BA} = \overrightarrow{OA} - \overrightarrow{OB} = \begin{pmatrix} -5 \\ 1 \end{pmatrix} - \begin{pmatrix} 2 \\ 0 \end{pmatrix} = \begin{pmatrix} -7 \\ 1 \end{pmatrix}$

$\begin{pmatrix} \cos 60° & -\sin 60° \\ \sin 60° & \cos 60° \end{pmatrix} \begin{pmatrix} -7 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{.2} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} -7 \\ 1 \end{pmatrix} = \begin{pmatrix} -\frac{7}{2} - \frac{\sqrt{3}}{2} \\ \frac{1}{2} - \frac{7\sqrt{3}}{2} \end{pmatrix}$

$\overrightarrow{OA'} = \overrightarrow{OB} + \begin{pmatrix} -\frac{7}{2} - \frac{\sqrt{3}}{2} \\ \frac{1}{2} - \frac{7\sqrt{3}}{2} \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \end{pmatrix} + \begin{pmatrix} -\frac{7}{2} - \frac{\sqrt{3}}{2} \\ \frac{1}{2} - \frac{7\sqrt{3}}{2} \end{pmatrix} = \begin{pmatrix} -\frac{3}{2} - \frac{\sqrt{3}}{2} \\ \frac{1}{2} - \frac{7\sqrt{3}}{2} \end{pmatrix}$

So the coordinates of $A'$ are $\left(-\frac{3}{2} - \frac{\sqrt{3}}{2}, \frac{1}{2} - \frac{7\sqrt{3}}{2}\right)$



**27)**

Let $P$ have coordinates $(a, b)$:

$\overrightarrow{PA} = \overrightarrow{OA} - \overrightarrow{OP} = \begin{pmatrix}0\\8\end{pmatrix} - \begin{pmatrix}a\\b\end{pmatrix} = \begin{pmatrix}-a\\8-b\end{pmatrix}$

$\begin{pmatrix}\cos 30° & -\sin 30°\\ \sin 30° & \cos 30°\end{pmatrix}\begin{pmatrix}-a\\8-b\end{pmatrix} = \begin{pmatrix}\frac{\sqrt{3}}{2} & -\frac{1}{2}\\ \frac{1}{2} & \frac{\sqrt{3}}{2}\end{pmatrix}\begin{pmatrix}-a\\8-b\end{pmatrix}$

$= \begin{pmatrix} -\frac{\sqrt{3}}{2}a - 4 + \frac{1}{2}b \\ -\frac{1}{2}a + 4\sqrt{3} - \frac{\sqrt{3}}{2}b \end{pmatrix}$

$\overrightarrow{OA'} = \overrightarrow{OP} + \begin{pmatrix} -\frac{\sqrt{3}}{2}a - 4 + \frac{1}{2}b \\ -\frac{1}{2}a + 4\sqrt{3} - \frac{\sqrt{3}}{2}b \end{pmatrix} = \begin{pmatrix}a\\b\end{pmatrix} + \begin{pmatrix} -\frac{\sqrt{3}}{2}a - 4 + \frac{1}{2}b \\ -\frac{1}{2}a + 4\sqrt{3} - \frac{\sqrt{3}}{2}b \end{pmatrix}$

$= \begin{pmatrix} \left(1 - \frac{\sqrt{3}}{2}\right)a - 4 + \frac{1}{2}b \\ -\frac{1}{2}a + 4\sqrt{3} + \left(1 - \frac{\sqrt{3}}{2}\right)b \end{pmatrix} = \begin{pmatrix}-5+\sqrt{3}\\3+3\sqrt{3}\end{pmatrix}$

$\left(1 - \frac{\sqrt{3}}{2}\right)a - 4 + \frac{1}{2}b = -5 + \sqrt{3} \Rightarrow \left(2 - \sqrt{3}\right)a - 8 + b = -10 + 2\sqrt{3}$

$\Rightarrow \left(2 - \sqrt{3}\right)a + b = -2 + 2\sqrt{3}$ (1)

$-\frac{1}{2}a + 4\sqrt{3} + \left(1 - \frac{\sqrt{3}}{2}\right)b = 3 + 3\sqrt{3} \Rightarrow -a + 8\sqrt{3} + \left(2 - \sqrt{3}\right)b = 6 + 6\sqrt{3}$

$\Rightarrow -a + \left(2 - \sqrt{3}\right)b = 6 - 2\sqrt{3}$ (2)

$\therefore a = -2$ and $b = 2$

So the coordinates of $P$ are $(-2, 2)$

